

Hand-visible analytic derivations for 103 finite-ledger predicates

1 Scope and logical convention

This note exposes, in hand-checkable form, the predicates not assigned to the other five ledger packets. They are precisely

11 threshold predicates + 48 zero predicates + 43 lower-closure predicates + 1 cap-74 predicate = 103.

The omitted lower-closure predicate is the single six-cap tuple identity; it belongs to the cap-inventory packet. Every inequality below is proved from a displayed analytic inequality, integer arithmetic, rational cross multiplication, or positive-side squaring. An executable ledger may check the same arithmetic, but no executable output is used as a premise.

We use

$$\omega_0 = \frac{\sqrt{3}}{2\pi} - \frac{1}{6}, \quad C_0 = 1 + \frac{8\sqrt{2}}{15}, \quad \rho_* = \frac{\omega_0}{2C_0}, \quad \rho_0 = \frac{1}{\sqrt{337}}, \quad \sigma = \frac{3\omega_0}{4}, \quad \rho_c = \frac{1}{1+2\pi}, \quad d = \frac{\sqrt{337}}{2}.$$

The elementary enclosure

$$\frac{333}{106} < \pi < \frac{355}{113} < \frac{22}{7} \tag{1}$$

is analytic: the lower and middle bounds follow from alternating truncations of Machin's identity $\pi = 16 \arctan(1/5) - 4 \arctan(1/239)$, and the last one is immediate by cross multiplication. For reference, 355/113 exceeds the upper Machin truncation by the positive rational numerator 45167474711 over denominator 189820334689453125. The classical integral

$$0 < \int_0^1 \frac{x^4(1-x)^4}{1+x^2} dx = \frac{22}{7} - \pi$$

also gives the final upper bound directly.

2 The eleven threshold predicates

First we obtain a rational two-sided enclosure for ω_0 . The exact integer residuals

$$3 \, 1101^2 113^2 - 607^2 355^2 = 1998482 > 0, \quad 25 \, 333^2 - 243 \, 106^2 = 41877 > 0$$

and (1) imply

$$\sqrt{3} > \frac{607 \, 355}{1101 \, 113} > \frac{607\pi}{1101}, \quad 9\sqrt{3} < 5\pi.$$

Consequently

$$\frac{40}{367} < \omega_0 < \frac{1}{9}. \tag{2}$$

Indeed, $607/2202 - 1/6 = 40/367$, while $5/18 - 1/6 = 1/9$.

Since $2 \cdot 32^2 - 45^2 = 23 > 0$, one has $\sqrt{2} > 45/32$, hence $C_0 > 7/4$ and $\rho_* < 2/63$. The following chain consists entirely of strict rational comparisons, with the indicated radical comparisons justified by squaring:

$$\rho_* < \frac{2}{63} < \frac{1}{\sqrt{337}} < \frac{1}{18} < \frac{3}{40} < \frac{3\omega_0}{4} < \frac{1}{12} < \frac{7}{51} < \frac{1}{1+2\pi} < \frac{1}{7} < \frac{39}{50} < \frac{7}{8}.$$

Here

$$4 \cdot 337 < 63^2, \quad 18^2 < 337, \quad 367 < 400, \quad 51 < 84, \quad 367 < 420,$$

and $7/51 < \rho_c < 1/7$ follows respectively from $\pi < 22/7$ and $\pi > 3$. This proves the five ratio-order rows. It also proves

$$\omega_0 < \frac{1}{9} < \frac{7}{51} < \rho_c.$$

Moreover

$$\omega_0 d > \frac{20\sqrt{337}}{367} > 1, \quad 400 \cdot 337 - 367^2 = 111 > 0.$$

The four terminal comparisons follow from the same bounds:

$$\begin{aligned} \sqrt{114} &< 18 < \frac{2}{\omega_0}, & \frac{2}{\omega_0} &< \frac{367}{20}, \\ \rho_c &< \frac{1}{7} < \frac{60}{367} < \frac{3\omega_0}{2}, & \omega_0 \sqrt{114} &> \frac{40\sqrt{114}}{367} > 1. \end{aligned}$$

The last positive-side squaring has residual

$$1600 \cdot 114 - 367^2 = 47711 > 0.$$

Thus the one-to-one threshold registry is

ID	source row	proved predicate
T1–T5	1–5	$\rho_* < \rho_0 < \sigma < \rho_c < 39/50 < 7/8$, one adjacent pair per row
T6	6	$\omega_0 < \rho_c$
T7	7	$\omega_0 d > 1$
T8	31	$\sqrt{114} < 2/\omega_0$
T9	32	$2/\omega_0 < 367/20$
T10	33	$\rho_c < 3\omega_0/2$
T11	34	$\omega_0 \sqrt{114} > 1$

3 The forty-eight zero specializations

3.1 A recurrence and elementary tangent bounds

Put

$$j_\ell(x) = \sqrt{\frac{\pi}{2x}} J_{\ell+1/2}(x), \quad P_\ell(x) = x^{\ell+1} j_\ell(x).$$

Then

$$P_{\ell+1} = (2\ell+1)P_\ell - x^2 P_{\ell-1}, \quad P_0 = \sin x, \quad P_1 = \sin x - x \cos x. \quad (3)$$

Write $P_\ell = A_\ell(t) \sin x + x C_\ell(t) \cos x$, $t = x^2$. The recurrence gives

$$\begin{aligned} A_2 &= 3 - t, & C_2 &= -3, \\ A_3 &= 15 - 6t, & C_3 &= t - 15, \\ A_5 &= 945 - 420t + 15t^2, & C_5 &= -945 + 105t - t^2, \\ A_6 &= 10395 - 4725t + 210t^2 - t^3, & C_6 &= -10395 + 1260t - 21t^2, \\ A_7 &= 135135 - 62370t + 3150t^2 - 28t^3, & C_7 &= -135135 + 17325t - 378t^2 + t^3. \end{aligned}$$

For $0 < s < \pi/2$, differentiation and integration give

$$\tan s > s, \quad \tan s > s + \frac{s^3}{3}. \quad (4)$$

For $0 < s < 1$, $\sin s < s$ and $\cos s > 1 - s^2/2$ give

$$\tan s < \frac{s}{1 - s^2/2}. \quad (5)$$

3.2 Sixteen half-period locations

For each internal face, the first displayed margin below is $w - (h_L/2)(22/7)$; the second is $(h_R/2)(333/106) - w$. Positivity therefore proves both strict half-period location rows without a decimal approximation.

(ℓ, n)	w	(h_L, h_R)	left margin	right margin
(1, 2)	77/10	(4, 5)	99/70	163/1060
(2, 2)	9	(5, 6)	8/7	45/106
(3, 2)	103/10	(6, 7)	61/70	737/1060
(5, 2)	129/10	(8, 9)	23/70	1311/1060
(1, 3)	21/2	(6, 7)	15/14	105/212
(2, 3)	61/5	(7, 8)	6/5	97/265
(6, 1)	10	(6, 7)	4/7	211/212
(7, 1)	23/2	(7, 8)	1/2	113/106

3.3 Sixteen half-period signs

At an integral or half-integral multiple of π , exactly one of $\sin x, \cos x$ vanishes. Direct substitution in the displayed polynomials gives the following complete sign table.

(ℓ, n)	h_L	left sign	(ℓ, n)	h_R	right sign
(1, 2)	4	—	(1, 2)	5	+
(2, 2)	5	—	(2, 2)	6	+
(3, 2)	6	—	(3, 2)	7	+
(5, 2)	8	—	(5, 2)	9	+
(1, 3)	6	+	(1, 3)	7	—
(2, 3)	7	+	(2, 3)	8	—
(6, 1)	6	+	(6, 1)	7	—
(7, 1)	7	+	(7, 1)	8	—

Here are explicit polynomial witnesses for the only signs not immediate from $A_1 = 1, C_1 = -1, A_2 = 3 - t, C_2 = -3, A_3 = 15 - 6t, C_3 = t - 15$. From (1), at $x = 4\pi$ one has $144 < t < 159$, at $x = 9\pi/2$ one

has $t > 182$, and at $x = 7\pi/2$ one has $110 < t < 121$. Monotonicity on these rational intervals and direct evaluation give

$$\begin{aligned} C_5(t) &< C_5(144) = -6561, & A_5(t) &> A_5(182) = 421365, \\ C_6(t) &< C_6(81) = -46116, & A_6(t) &> A_6(110) = 700645, \\ A_7(t) &< A_7(110) = -5878565, & C_7(t) &< C_7(144) = -2492559. \end{aligned}$$

For example, $A'_6 = -4725 + 420t - 3t^2$ is positive on $[110, 121]$, while A'_7 is negative there; the other asserted monotonicities follow at once from their displayed derivatives. Combining these signs with the quadrant signs of sine and cosine proves all sixteen entries.

3.4 Eight rational-wall signs

Substitution in (3) reduces each rational-wall sign to one of the following exact tangent comparisons. The shifted angle is always in $(0, \pi/2)$, as already proved by the location table.

(ℓ, n)	wall	hand certificate	$\text{sgn } j_\ell(w)$
(1, 2)	77/10	For $u = 5\pi/2 - w$, $\tan u > 3/20 > 10/77 = 1/w$; hence $\tan(w - 2\pi) = \cot u < w$.	−
(2, 2)	9	For $u = 3\pi - w$, $\tan u > 21/50 > 9/26 = 3w/(w^2 - 3)$.	−
(3, 2)	103/10	For $u = 7\pi/2 - w$, $\tan u > 69/100 > 621540/938227 = (6w^2 - 15)/(w(w^2 - 15))$.	−
(5, 2)	129/10	For $y = w - 4\pi$, $0 < y < 17/50$, so $\tan y < 1700/4711 < 177800829/427700150 = -wC_5/A_5$.	−
(1, 3)	21/2	For $u = 7\pi/2 - w$, $\tan u > 49/100 > 2/21 = 1/w$, hence $\tan(w - 3\pi) < w$.	+
(2, 3)	61/5	For $u = 4\pi - w$, $\tan u > 9/25 > 915/3646 = 3w/(w^2 - 3)$.	+
(6, 1)	10	For $y = w - 3\pi$, $4/7 < y < 29/50$, so $\tan y < 2900/4159 < 188790/127579 = -wC_6/A_6$.	+
(7, 1)	23/2	For $u = 4\pi - w$, $\tan u > u + u^3/3 > 546377/375000 > 3153151489/2276518664 = wC_7/A_7$.	+

For complete cross-multiplication visibility, the four less immediate positive gaps in that table are

$$\begin{aligned} \frac{69}{100} - \frac{621540}{938227} &= \frac{2583663}{93822700}, \\ \frac{177800829}{427700150} - \frac{1700}{4711} &= \frac{110529450419}{2014895406650}, \\ \frac{188790}{127579} - \frac{2900}{4159} &= \frac{415198510}{530601061}, \\ \frac{546377}{375000} - \frac{3153151489}{2276518664} &= \frac{3837851856583}{53355906187500}. \end{aligned}$$

The four small gaps are respectively $31/1540$, $24/325$, $829/2100$, $9939/91150$.

3.5 Intermediate rational-wall signs used by the index induction

The coefficient signs make all unmentioned entries in the sign vectors immediate. The remaining intermediate entries require the following exact thresholds. At $w = 10$, with $y = 10 - 3\pi$,

$$\tan y < \frac{2900}{4159} < \frac{170}{117} < 10, \quad \tan y > y > \frac{4}{7} > \frac{890}{21789}.$$

Substitution in P_1, P_3, P_5 gives $\text{sgn}(j_1, j_3, j_5) = (+, -, -)$, and $P_4 = 7P_3 - w^2P_2 < 0$. At $w = 103/10$, with $u = 7\pi/2 - w$,

$$\tan u > \frac{69}{100} > \frac{10}{103},$$

which gives $j_1(w) > 0$. At $w = 23/2$, with $u = 4\pi - w$,

$$\tan u > \frac{546377}{375000} > \frac{3153151489}{2276518664} > \frac{224020}{186301} > \frac{138}{517}.$$

Besides the target j_7 sign, the last two thresholds give $j_4(w) < 0$ and $j_2(w) > 0$; the recurrence gives the intervening signs. Finally, at $w = 129/10$, with $y = w - 4\pi$,

$$\tan y < \frac{1700}{4711} < \frac{177800829}{427700150} < \frac{651063}{327820} < \frac{129}{10}.$$

Besides the target j_5 sign, this proves $j_3(w) > 0$ and $j_1(w) < 0$, and the recurrence gives the remaining signs. Every sign used below is therefore supported by a displayed rational comparison.

3.6 Exact zero indices from adjacent-order interlacing

For $x > 0$, define the strict positive-zero count

$$Z_\ell(x) := \#\{n \in \mathbb{N} : j_{\ell+1/2,n} < x\}.$$

Lemma 3.1 (Interlacing-to-index principle). *For every $\ell \in \mathbb{N}_0$ and $n \in \mathbb{N}$,*

$$j_{\ell+1/2,n} < j_{\ell+3/2,n} < j_{\ell+1/2,n+1}. \quad (6)$$

If x is not a zero of $j_0, \dots, j_L$, then, for $0 \leq \ell < L$,

$$Z_{\ell+1}(x) \in \{\max\{Z_\ell(x) - 1, 0\}, Z_\ell(x)\}, \quad \text{sgn } j_{\ell+1}(x) = (-1)^{Z_{\ell+1}(x)}. \quad (7)$$

Consequently $Z_0(x)$, together with the signs of $j_1(x), \dots, j_L(x)$, determines all the counts $Z_1(x), \dots, Z_L(x)$ uniquely.

Proof. DLMF 10.21.2 states, for $\nu \geq 0$,

$$j_{\nu,1} < j_{\nu+1,1} < j_{\nu,2} < j_{\nu+1,2} < j_{\nu,3} < \dots$$

Taking $\nu = \ell + 1/2$ proves (6). Put $q = Z_\ell(x)$. The first inequality in (6) shows that at most q zeros of order $\ell + 3/2$ lie below x . If $q \geq 1$, the second inequality shows that its first $q - 1$ zeros lie below $j_{\ell+1/2,q} < x$. Thus

$$\max\{q - 1, 0\} \leq Z_{\ell+1}(x) \leq q,$$

which is the first assertion in (7).

The spherical Bessel function j_ℓ is positive for all sufficiently small positive arguments. Its positive zeros are simple by uniqueness for the Bessel equation, so its sign changes at every positive zero. This proves the parity identity in (7). When $q \geq 1$, the two possible counts have opposite parity; when $q = 0$, only zero is possible. The sign therefore selects the count uniquely. $\square$

Because $j_0(x) = \sin x/x$, one has $j_{1/2,n} = n\pi$. The strict locations in the first table therefore give $Z_0(w)$ exactly. Applying Lemma 3.1 successively to the nonzero sign vectors obtained from (3) and the exact tangent comparisons above gives:

wall w	target (ℓ, n)	location of w	$Z_0(w)$	$\text{sgn}(j_0, \dots, j_\ell)(w)$	$(Z_0, \dots, Z_\ell)(w)$
77/10	(1, 2)	$(2\pi, 5\pi/2)$	2	(+, -)	(2, 1)
9	(2, 2)	$(5\pi/2, 3\pi)$	2	(+, +, -)	(2, 2, 1)
10	(6, 1)	$(3\pi, 7\pi/2)$	3	(-, +, +, -, -, +)	(3, 2, 2, 1, 1, 0)
103/10	(3, 2)	$(3\pi, 7\pi/2)$	3	(-, +, +, -)	(3, 2, 2, 1)
21/2	(1, 3)	$(3\pi, 7\pi/2)$	3	(-, +)	(3, 2)
23/2	(7, 1)	$(7\pi/2, 4\pi)$	3	(-, -, +, +, -, -, +)	(3, 3, 2, 2, 1, 1, 0)
61/5	(2, 3)	$(7\pi/2, 4\pi)$	3	(-, -, +)	(3, 3, 2)
129/10	(5, 2)	$(4\pi, 9\pi/2)$	4	(+, -, -, +, +, -)	(4, 3, 3, 2, 2, 1)

In every row the target component is $Z_\ell(w) = n - 1$, and the displayed sign is nonzero. Hence, with the convention $j_{\nu,0} = 0$,

$$j_{\ell+1/2, n-1} < w < j_{\ell+1/2, n}.$$

Thus the unique next zero is the claimed n -th zero, and the eight rows prove

$$\begin{aligned} j_{3/2,2} &> 77/10, & j_{5/2,2} &> 9, & j_{7/2,2} &> 103/10, & j_{11/2,2} &> 129/10, \\ j_{3/2,3} &> 21/2, & j_{5/2,3} &> 61/5, & j_{13/2,1} &> 10, & j_{15/2,1} &> 23/2. \end{aligned} \quad (8)$$

3.7 Six Lorch rows and two wall-order rows

For $\nu = \ell + 1/2$, the second strict inequality, item (ii), displayed in the published abstract of Lorch's article *Some inequalities for the first positive zeros of Bessel functions*, SIAM J. Math. Anal. **24** (1993), 814–823, doi:10.1137/0524050, is valid for $-1 < \nu < \infty$ and states

$$j_{\nu,1}^2 > \frac{24(\nu+1)^2}{1-2\nu+\sqrt{(2\nu+3)(2\nu+11)}} - 2(\nu^2-1) =: R_\ell.$$

After rationalization,

$$R_\ell = \frac{3(2\ell+3)\sqrt{(\ell+2)(\ell+6)} - 2\ell^2 + \ell + 6}{4}.$$

All sides being squared are positive, and the following integer residuals prove the three required strict specializations:

ℓ	wall c	positive squared residual proving $R_\ell > c^2$
8	71/6	$1026^2 \cdot 35 - 6067^2 = 35171$
9	64/5	$1575^2 \cdot 165 - 20059^2 = 6939644$
10	69/5	$3450^2 \cdot 3 - 5911^2 = 767579$

Since $j_\ell(x) > 0$ for sufficiently small $x > 0$, and its first positive zero is strictly to the right of c , these same three rows imply $j_\ell(c) > 0$. Thus the three executable sign rows are analytic corollaries, not independent numerical assumptions. Finally,

$$\frac{103}{10} - \frac{81}{8} = \frac{7}{40} > 0, \quad \frac{61}{5} - \frac{21}{2} = \frac{17}{10} > 0.$$

The one-to-one zero registry is as follows. Number the eight internal faces in the order of the location table. For face r , IDs $Z_{5r-4}, \dots, Z_{5r}$ are, in order: right of the left half-period, left of the right half-period, left half-period sign, right half-period sign, and rational-wall sign. This bijects IDs Z1–Z40 with the first forty source rows. The remaining bijection is

ID	source row	predicate
Z41	41	Lorch specialization $\ell = 8, c = 71/6$
Z42	42	positive sign at $\ell = 8, c = 71/6$
Z43	43	Lorch specialization $\ell = 9, c = 64/5$
Z44	44	positive sign at $\ell = 9, c = 64/5$
Z45	45	Lorch specialization $\ell = 10, c = 69/5$
Z46	46	positive sign at $\ell = 10, c = 69/5$
Z47	47	$103/10 > 81/8$
Z48	48	$61/5 > 21/2$

4 The forty-three lower-closure predicates

4.1 Eight activation and exclusion rows

The complete arithmetic is

$$\begin{aligned}
(23/2)^2 - 114 &= 73/4 > 0, & 10^2 &= 100, \\
(129/10)^2 - 114 &= 5241/100 > 0, & 103/10 - 81/8 &= 7/40 > 0, \\
(81/8)^2 + 8 &= 7073/64, & (81/8)^2 + 18 - 114 &= 417/64 > 0, \\
(21/2)^2 + 4 - 114 &= 1/4 > 0, & 16\pi^2 &> 16 \cdot 3^2 > 114.
\end{aligned}$$

These are exactly lower IDs L1–L8, corresponding to source rows 1–8.

4.2 Five Weyl payments

For $0 < \rho < \rho_c < 1/7$, (1) yields the uniform analytic lower bound

$$W(\rho, K) = \frac{2}{9\pi}(1 - \rho^3)K^3 > \frac{38}{539}K^3. \quad (9)$$

Using $d > 9$ and $\sqrt{7073}/8 > 21/2$, the five payments are the following exact positive reserves:

ID	source row	threshold/cap	reserve from (9)
L9	10	$K > d, 46$	$(38/539)9^3 - 46 = 2908/539$
L10	11	$K > 10, 59$	$(38/539)10^3 - 59 = 6199/539$
L11	12	$K > 81/8, 66$	$(38/539)(81/8)^3 - 66 = 990435/137984$
L12	13	$K > 21/2, 69$	$(38/539)(21/2)^3 - 69 = 555/44$
L13	14	$K > \sqrt{7073}/8, 78$	$(38/539)(21/2)^3 - 78 = 159/44$

4.3 The cap–395 cell: eighteen rows

Put $R = 367/20$, $z_\ell = \ell + 1/2$. The analytic turning estimate

$$G_K(z) < \frac{(K - z)^{3/2}}{3\sqrt{K}}$$

is increasing in K , so on $K < R$ a strict cap c_ℓ follows from

$$\frac{(R - z_\ell)^3}{9R} = \frac{(357 - 20\ell)^3}{1321200} < \left(c_\ell + \frac{3}{4}\right)^2. \quad (10)$$

The following table gives all fifteen caps and the exact positive numerator $82575(4c_\ell + 3)^2 - (357 - 20\ell)^3$ obtained by clearing the positive denominators in (10).

ℓ	0	1	2	3	4	5	6	7
c_ℓ	6	5	5	4	4	3	3	3
margin	14697882	5409422	11827162	3611502	8555642	1604782	5267322	8361062

ℓ	8	9	10	11	12	13	14
c_ℓ	2	2	1	1	1	1	0
margin	2346202	4446342	176282	1474822	2444562	3133502	286642

Thus L15–L29 correspond bijectively to source rows 16–30, with L(15 + ℓ) the ℓ -th table column. Since $G_R(z)$ is strictly decreasing in z , the $c_{14} = 0$ row removes every $\ell \geq 14$ tail; this is L30/source row 31.

The weighted cap itself, L14/source row 15, is visible from the contributions

$$6, 15, 25, 28, 36, 33, 39, 45, 34, 38, 21, 23, 25, 27, \quad \sum_{\ell=0}^{13} (2\ell + 1)c_\ell = 395.$$

Finally, on the exceptional cell $\rho K \geq 5/2$, with $\rho < 11/80$ and $1/\pi > 113/355$,

$$\begin{aligned} W(\rho, K) &> \frac{2}{9} \frac{113}{355} \left(\frac{5}{2}\right)^3 \left[\left(\frac{80}{11}\right)^3 - 1 \right] \\ &= \frac{160293325}{378004} = 395 + \frac{10981745}{378004} > 395. \end{aligned}$$

This is L31/source row 32.

4.4 Twelve floor and boundary-owner rows

The nine finite arithmetic rows are instances of one identity:

$$\frac{3}{2}(p+1) \leq 2p + \frac{1}{2} \quad (p \geq 2), \quad \left(2p + \frac{1}{2}\right) - \frac{3}{2}(p+1) = \frac{p-2}{2} \geq 0. \quad (11)$$

At $p = 2$, $(3/2)3 = 9/2$; this is L32/source row 33. For $p = 2, \dots, 9$, formula (11) is respectively L33–L40/source rows 34–41.

The half-integer face is exact: $5/2 = 2 + 1/2$. Assigning it to the high side makes the interface remainder vanish and leaves the exceptional cell closed; this is L41/source row 42. At $K = 2/\omega_0$,

$$p = \lfloor \omega_0 K \rfloor = 2, \quad \rho K \leq \frac{2\rho_c}{\omega_0} < 3$$

by T10. This proves L42/source row 43. Moreover

$$\rho K - \frac{1}{2} < \frac{5}{2}, \quad m = \left\lfloor \rho K - \frac{1}{2} \right\rfloor \leq 2 \leq 2p - 1 = 3,$$

and the strict exceptional condition $K < 2/\omega_0$ fails at equality. Thus the complementary average-floor branch owns the equality face, proving L43/source row 44. More generally, if $K \geq 2/\omega_0$, then $p \geq 2$ and

$$\rho K < \frac{3}{2}(p+1) \leq 2p + \frac{1}{2}, \quad m \leq 2p - 1,$$

so the finite instances conceal no unproved large- p case.

5 The supplemental cap-74 incompatibility

The propagated ball wall requires $z^2 > 2409/100$, while the product wall requires $z^2 < 70/3$. They are incompatible because

$$\frac{2409}{100} - \frac{70}{3} = \frac{7227 - 7000}{300} = \frac{227}{300} > 0.$$

This is supplemental ID S1 and is a hand proof, not merely a reported certificate value.

6 Census and source-visibility verdict

family	count	IDs	hand-visible status before this note
threshold remainder	11	T1–T11	printed, with a few implications compressed
zero specializations	48	Z1–Z48	tables printed; several recurrence steps compressed
lower remainder	43	L1–L43	payments printed; floor instances generated implicitly
supplemental cap-74	1	S1	printed exactly
total	103		

No predicate in this 103-row allocation is an unsupported executable-only mathematical premise. Before this note, the principal visibility gaps were the one-to-one expansion of the forty zero sign/location rows, the fact that the three Lorch-wall signs are logical corollaries of the strict first-zero bounds, and the expansion of the twelve floor/boundary rows. The displayed recurrences, rational margins, general floor identity, and exact ownership arguments close those presentation gaps.